

Forward Pricing

What should be the fair price for a forward contract? We can answer this question by considering the costs and benefits of buying now compared with buying on some future date. In a forward contract, the costs and benefits are not eliminated; they are simply deferred. They should therefore be reflected in the forward price.

$$\text{forward price} = \text{current cash price} + \text{costs of buying now} \\ - \text{benefits of buying now}$$

Let's return to our example from Chapter 1 where my friend Jerry wanted to acquire land on which to build a restaurant. He was considering both a cash purchase and a one-year forward contract. If he enters into a forward contract, what should be a fair one-year forward price for the land?

If Jerry wants to buy the land right now, he will have to pay Farmer Smith's asking price of \$100,000. However, in researching the feasibility of a one-year forward contract, Jerry has learned the following:

1. The cost of money, whether borrowing or lending,¹ is currently 8.00 percent annually.
2. The owner of the land must pay \$2,000 in real estate taxes; the taxes are due in nine months.
3. There is a small oil well on the land that pumps oil at the rate of \$500 per month; the oil revenue is receivable at the end of each month.

If Jerry decides to buy the land now, what are the costs compared with buying the land one year from now? First, Jerry will have to borrow \$100,000 from the local bank. At a rate of 8 percent, the one-year interest costs will be

$$8\% \times \$100,000 = \$8,000$$

¹ At this point, we will assume that the same interest rate applies to all transactions, whether borrowing or lending. Admittedly, for a trader, the interest cost of borrowing will almost always be higher than the interest earned when lending.